# Mohammad Khamaysi

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## SUMMARY

Digital and mixed-signal ASIC designer, M.Sc. student at Tel Aviv University. Designed and signed off **two chips in TSMC 65 nm LP**, submitted for fabrication in September 2026 through a multi-project wafer. Owned the whole flow from a MATLAB golden model to GDSII, with a bit-exact compare against that model at every stage. Research interests: energy-efficient DSP hardware, multiplierless arithmetic, digital front ends for transmitters, algorithm-to-silicon co-design.

## EDUCATION

**M.Sc. Electrical and Electronic Engineering**, Tel Aviv University, GPA 93 2024 – present

Excellence Scholarship 2026. Thesis: *Low-Power Arithmetic for Configurable Multi-Rate Digital Interpolation Filters, from MATLAB to GDSII: Polyphase, Cubic and Spline Compared on Two TSMC 65 nm Tapeouts*, supervisor Dr. Nimrod Ginzberg; draft completed October 2026.

**B.Sc. Electrical and Electronic Engineering**, Tel Aviv University, GPA 84 2020 – 2024

Courses: VLSI Verification & Synthesis (100), Mixed-Signal Circuits (99), Random Signals and Noise (95), Introduction to VLSI (94), Integrated Analog IC Design (94), Digital VLSI Design (89), Signals and Systems (89).

## RESEARCH: TWO INTERPOLATION-FILTER ASICS, MATLAB TO GDSII

- **Two tapeouts.** Chip 1 with loadable filter coefficients, chip 2 with every coefficient hardwired, its polyphase and cleanup filters built from shift-add (signed-power-of-two) trees with no multiplier. 0.96 × 0.96 mm die, 960 MHz, seven run-time modes: three word lengths, four interpolation factors.
- **Three interpolators per die**, polyphase FIR, cubic and MinAJ2 Hermite spline, behind one control unit and one cleanup filter; equiripple band edges chosen per interpolation factor, 7.7 dB more image rejection at the same EVM and zero hardware cost.
- **Multiplier-free arithmetic.** Removing the multipliers from the polyphase and cleanup datapaths cut the polyphase core by 40 % in area and in power (16.6 to 9.8 mW at 960 MHz, L = 4) at the same EVM; across all cores chip 2 draws 39 to 60 % less power and uses 20 % less cell area.
- **Bit-exact verification.** One golden fixed-point model compared word for word at RTL, gate, pad-ring and post-layout SDF level; 18 of 18 post-layout runs bit-exact at 720, 900 and 960 MHz at both corners. Block tests, functional coverage and VC Formal proofs.
- **Physical design and sign-off.** Innovus floorplan, power grid, placement, CCOpt clock tree, routing; Voltus power and IR drop; PrimeTime three-corner sign-off; Conformal RTL-to-gate and gate-to-gate equivalence; Calibre DRC, LVS, ERC; bond pads, dummy fill and final GDS in Virtuoso.
- **Signal quality.** EVM as low as 54 m% rms on the polyphase core, read on a Keysight 89600 vector signal analyser; power from Voltus on the sign-off database with the activity of the bit-exact simulation.

## TEACHING AND SUPERVISION

- Tutor: Digital Electronic Circuits, Physics II, Introduction to VLSI. Teaching assistant: Energy Conversion Laboratory, Tel Aviv University.
- Supervised four undergraduate students in their final ASIC design projects.

## SKILLS

- **Languages:** SystemVerilog, Verilog, MATLAB, Python, C, Tcl.
- **Tools:** Synopsys Design Compiler, VCS, Verdi, VC Formal, PrimeTime; Cadence Innovus, Voltus, Conformal, Virtuoso and Spectre; Siemens Calibre; Keysight 89600 VSA.
- **Topics:** multi-rate DSP and fixed-point design, multiplierless (SPT/CSD) arithmetic, clock gating, timing closure, multi-corner sign-off, power and IR analysis, QAM transmitter chains.

## COURSE PROJECTS

- 6-bit 2 GHz differential flash ADC in Virtuoso, transistor level, with INL/DNL, SNDR/ENOB and power analysis.
- Fully differential high-speed OTA with common-mode feedback; 4-bit ALU in a full VLSI flow with timing, area and power.

## PERSONAL

Oud player since 2018; builds electronic musical instruments; member of the Tel Aviv University oriental music band; swimming. Languages: Arabic (native), Hebrew (fluent), English (professional).